8.2 Transformations of Logarithmic Functions

A Transformations of Logarithmic Functions

The function:

$$g(x) = A\log_b B(x-C)+D$$

is a transformation of the parent function $f(x)=\log_b x$.

Here are some features of the function $g(x)$:

Domain:

- If $C>0$ then $D=(C,\infty)$
- If $C<0$ then $D=(-\infty,C)$

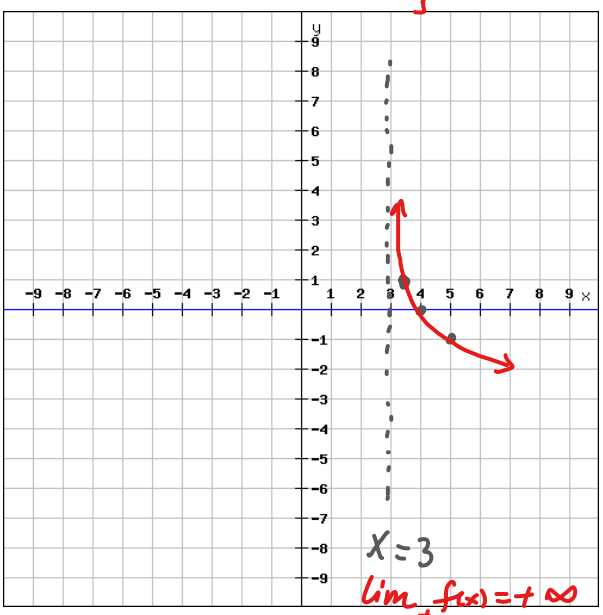
Range: R

Vertical Asymptote: $x=C$

Ex 1. For each case, use three key points to graph the logarithmic function. Specify the domain, range, and the equation of the vertical asymptote.

a) $y=-\log_2(x-3)$

$D=\{x\in R / x>3\}$
$R=\{y\in R\}$



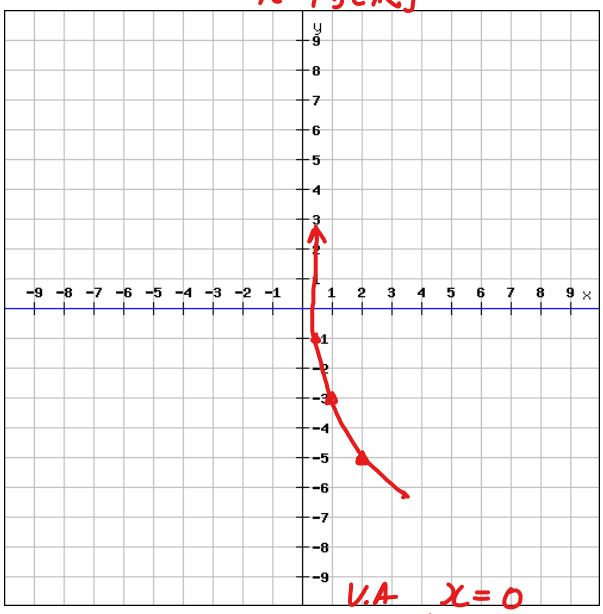
$x=3$

$\lim_{x\to 3^+} f(x)=+\infty$

$(\frac{1}{2},-1)\to(\frac{7}{2},1)$
$(1,0)\to(4,0)$
$(2,1)\to(5,-1)$

b) $y=2\log_{0.5}x-3$

$D=\{x\in R / x>0\}$
$R=\{y\in R\}$



V.A $x=0$

$\lim_{x\to 0^+} f(x)=+\infty$

$(2,-1)\to(2,-5)$
$(1,0)\to(1,-3)$
$(\frac{1}{2},1)\to(\frac{1}{2},-1)$

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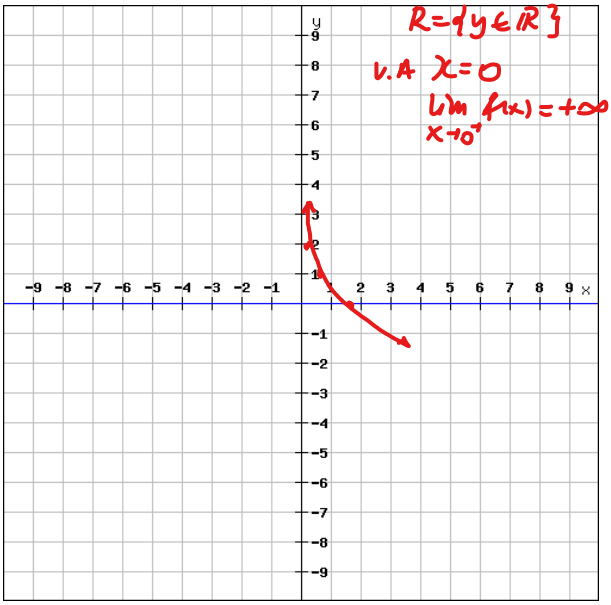
c) $y = 1 - \log_3(2x)$ $= -\log_3 2x + 1$

$D = \{x \in \mathbb{R} / x > 0\}$

$R = \{y \in \mathbb{R}\}$

V.A $x = 0$

$\lim_{x \to 0^+} f(x) = +\infty$



$(\frac{1}{3}, -1) \longrightarrow (\frac{1}{6}, 2)$

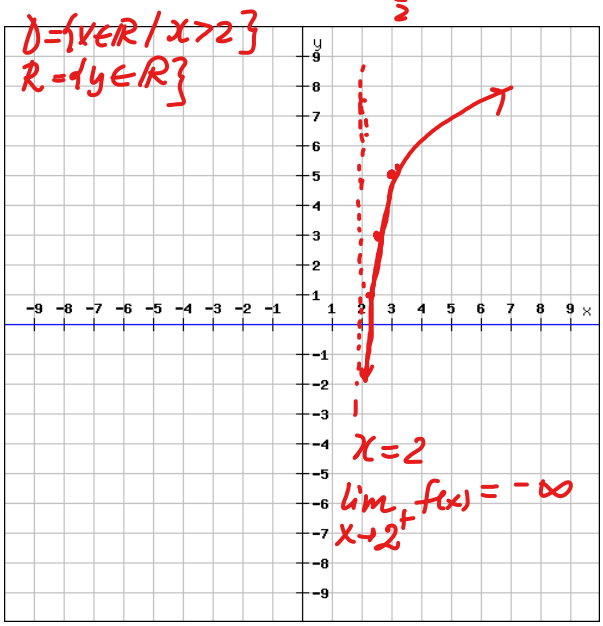
$(1, 0) \longrightarrow (\frac{1}{2}, 1)$

$(3, 1) \longrightarrow (\frac{3}{2}, 0)$

d) $y = -2\log_{0.5}(2x - 4) + 3$ $= -2\log_{\frac{1}{2}} 2(x-2) + 3$

$D = \{x \in \mathbb{R} / x > 2\}$

$R = \{y \in \mathbb{R}\}$



$x = 2$

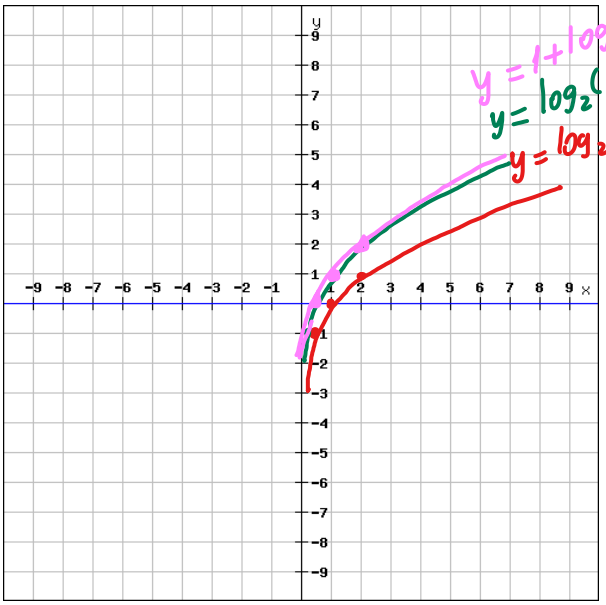
$\lim_{x \to 2^+} f(x) = -\infty$

$(\frac{1}{2}, 1) \longrightarrow (\frac{9}{4}, 1)$

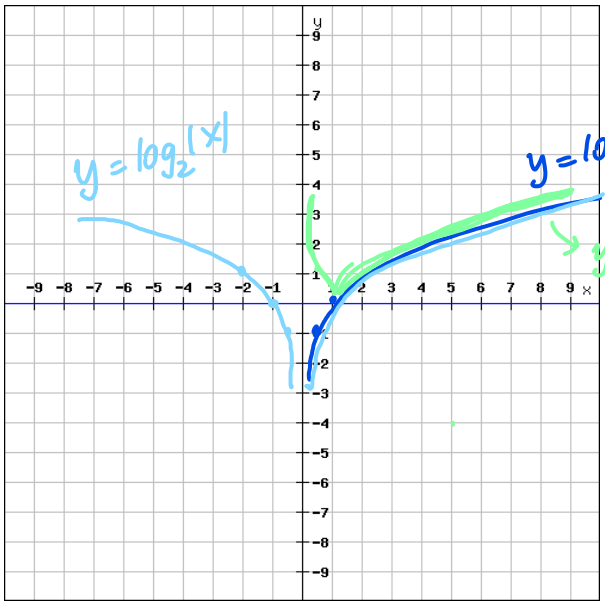
$(1, 0) \longrightarrow (\frac{5}{2}, 3)$

$(2, -1) \longrightarrow (3, 5)$

Ex 2. Graph on the same grid $y = \log_2 x$, $y = \log_2(2x)$, and $y = 1 + \log_2 x$.



Ex 3. Graph on the same grid $y = \log_2 x$, $y = \log_2 |x|$, and $y = |\log_2 x|$.



Reading: Nelson Textbook, Pages 452-457

Homework: Nelson Textbook, Page 457: #4, 5, 6, 7, 8, 9, 11

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